JCSSE 2026 · STREAMING ML · EXPLAINABLE AI

Enhanced Adversarial Drift Detection

for MLOps Feature Monitoring

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EADD answers WHETHER, WHEN, WHICH, and **HOW SEVERE** — in one pipeline.

01

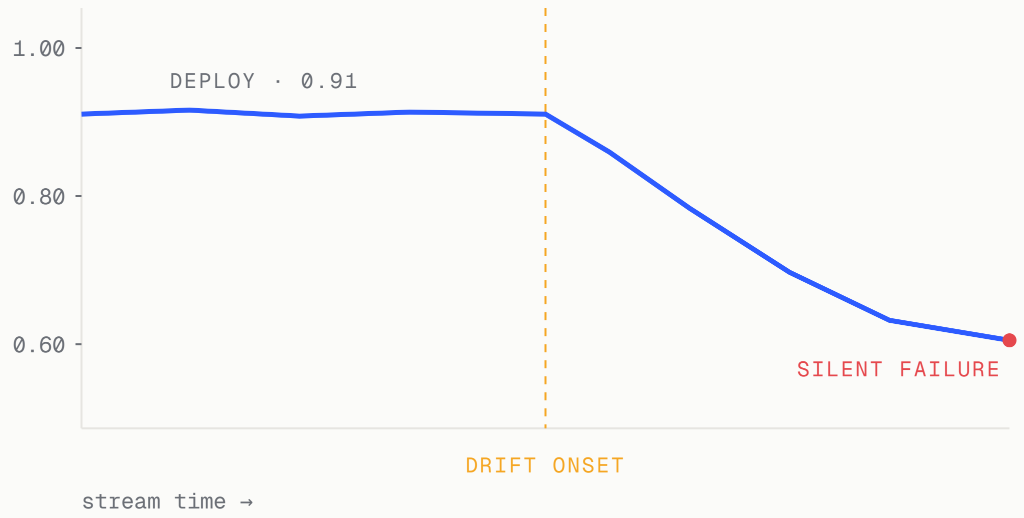
SECTION 01

The Problem

Why production ML detectors miss the mark.

Production ML fails silently

ACCURACY AFTER DRIFT ONSET



Which features? How severe? Real or noise?

D3 — GÖZÜAÇIK ET AL., CIKM 2019

Most robust unsupervised detector

Ranked most robust in **Lukats et al. 2024** — yet three structural gaps remain.

01 **No formal significance test** → false alarms on autocorrelated streams.

02 **No feature attribution** → "detect-and-discard" classifier.

03 **Linear classifier** → misses non-linear / incremental drift.

Capability matrix

DETECTOR	WHETHER	WHICH	SEVERITY	STREAMING
DDM / EDDM	✓	✗	✗	✓
KS / KSWIN	✓	~	✗	✓
D3	✓	✗	✗	✓
Vertex AI / Clarify	✓	✓	~	✗
TRIPODD / HDPD	✓	✓	✗	✗
EADD (this work)	✓	✓	✓	✓

> **No prior unsupervised streaming detector integrates whether + which + severity.**

✓ supported ~ partial ✗ none

Three contributions

01

ASPT

ADAPTIVE SEQUENTIAL PERMUTATION TEST

A formal permutation test gates drift confirmation — restoring statistical rigour to unsupervised detection.

90-95% fewer permutations · confirms at $i=20$ vs $B_{max}=199$ · inspired by Gandy 2009

02

SHAP + DSI

ATTRIBUTION & SEVERITY

TreeSHAP on the C2ST classifier identifies *which* features drifted; an entropy-normalised index scores *how severe*.

$DSI = AUC \cdot (1 - H_{norm}(s))$

03

Benchmark

VS 7 UNSUPERVISED DETECTORS

EADD evaluated against 7 detectors on synthetic streams + 13 real-world streams.

4/4 coverage · 0 false alarms · P@k = R@k = 1.0 (univariate)

02

SECTION 02

The EADD Pipeline

Four-step streaming detector.

End-to-end pipeline

STREAM
$\{x_t\} \in \mathbb{R}^d$ →

① ADAPTIVE WINDOWS
W_{ref} reservoir · W_{cur} sliding →

② LIGHTGBM C2ST
AUC · H_{pred} →

◇ AUC > τ ?

↳ yes →

③ ASPT
sequential perm. test →

◇ p < α ? →

④ TreeSHAP + DSI
attribution · severity →

DRIFT ALERT
features · severity

NO ×
Either gate fails → no drift · resume stream

Two-stage gate ⇒ Type I error bound: $\Pr(\mathrm{AUC} > \tau \mid H_0) \cdot \alpha \leq \alpha$

Adaptive Sequential Permutation Test

PROBLEM → IDEA

D3's fixed AUC threshold → false alarms on autocorrelated streams.

Idea: permute labels iteratively; stop early when the evidence is decisive.

Early reject H_0 — confirm drift

$c_i=0,\quad i \geq B_{min},\quad 1/(i+1) < \alpha \quad \Rightarrow \quad p = 1/(i+1)$

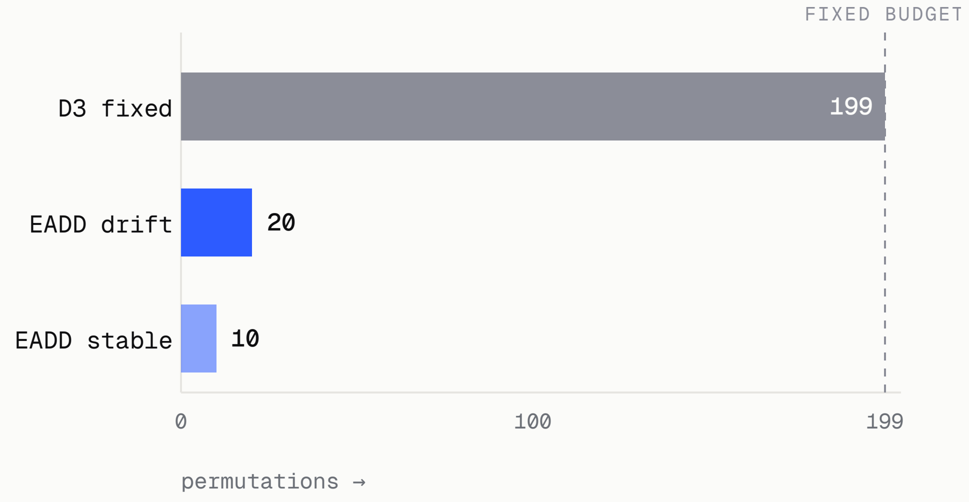
Early accept H_0 — no drift

$c_i/i > 2\alpha,\quad i \geq B_{min} \quad \Rightarrow \quad p = (c_i+1)/(i+1)$

Terminal

$p = (c_{B_{max}}+1)/(B_{max}+1)$

PERMUTATIONS USED PER DECISION



90-95% fewer permutations · α preserved

SHAP attribution & severity

WHICH FEATURES

$\bar{\varphi}_j = (1/n)\ \Sigma_i\ |\varphi_j^{(i)}|$

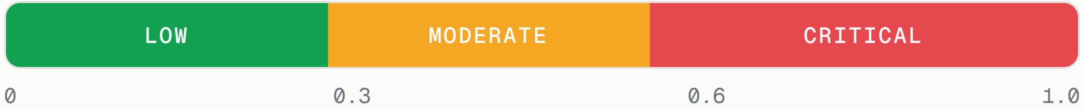
$s_j = \bar{\varphi}_j\ /\ \Sigma_k\ \bar{\varphi}_k$

$$\boxed{\mathrm{DSI} = \mathrm{AUC} \times (1 - H_{norm}(s))}$$

$H_{norm}(s) = -\Sigma_j\ s_j \log_2 s_j\ /\ \log_2 d \quad \in [0,1]$

Dimensionality-invariant — comparable across feature counts.

SEVERITY GRADIENT



Low <0.3	Scheduled retrain
Moderate 0.3–0.6	Audit shared source
Critical >0.6	Audit pipeline

Heuristic tiers; require domain calibration.

03

SECTION 03

Experiments & Results

What does EADD actually do?

Experimental setup

EXP 1

Synthetic

- n=5,000, d=5
- 4 drift patterns: abrupt / gradual / incremental / recurring
- 4 stable streams for false-alarm robustness

EXP 2

Real-world

- 13 streams · Lukats+ 2024
- 4 annotated INSECTS · 9 unannotated
- Metrics: Precision / Recall / MTD

EXP 3

SHAP accuracy

- n=10,000, d=10; drift @ t=5,000
- univariate / subset / multivariate
- Metrics: P@k / R@k / DSI

BASELINES

D3 · BNDM · CSDDM · IBDD · OCDD · SPLL · UDetect

CONFIG

LightGBM (50 trees) · N_{ref}=500 · N_{cur}=200 · M=50 · B_{max}=199 · B_{min}=10 · α=0.05 · τ=0.7

What EADD delivers

4/4

SYNTHETIC DRIFT COVERAGE

0

FALSE ALARMS ON STABLE STREAMS

8×

FASTER MTD VS D3 (INCREMENTAL)

1.00

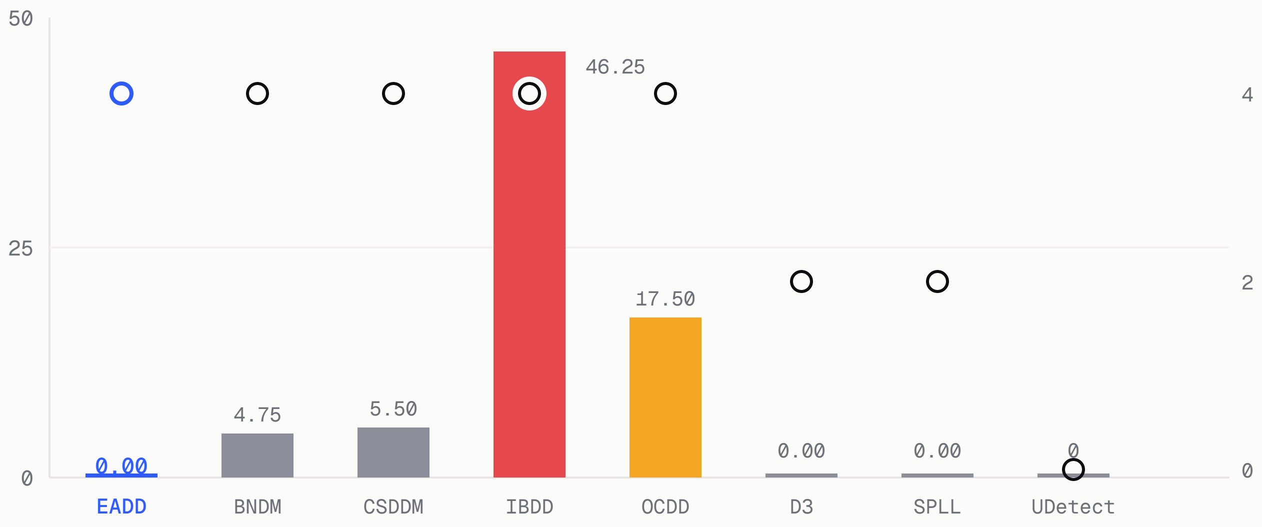
P@K = R@K (UNIVARIATE SHAP)

EADD is the only detector achieving **both** full drift-type coverage **and** zero false alarms — while answering *which* and *how severe*, in one streaming pipeline.

Mann-Whitney U vs D3 on autocorrelated streams: $p = 0.0101$ ($\tau = 0.6$).

Coverage vs false alarms

MEAN FALSE ALARMS PER TYPE (BARS) · COVERAGE /4 (○)



bars = mean FA · ○ = coverage (right axis)

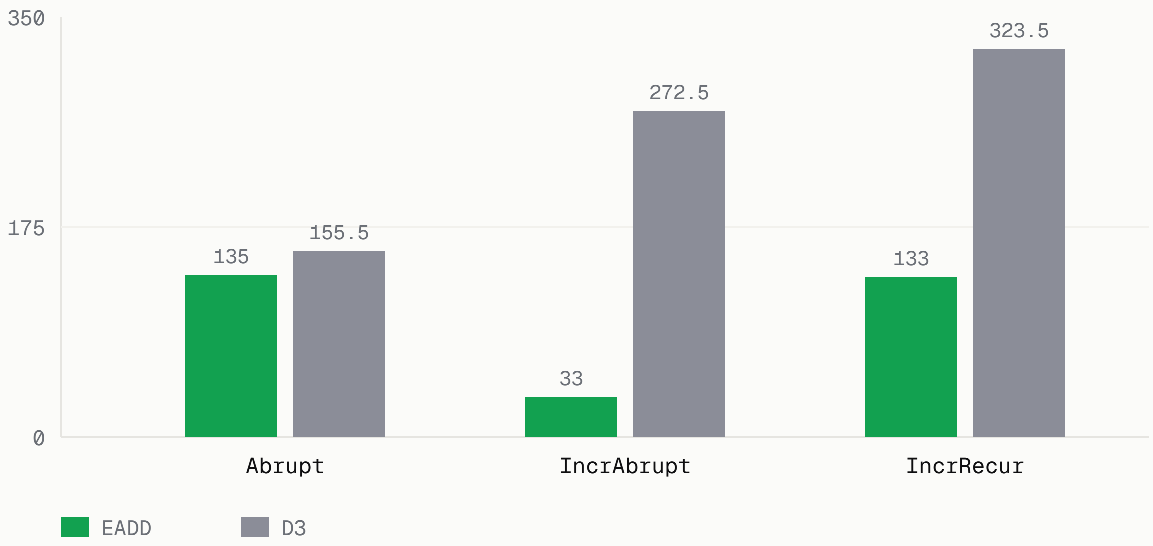
EADD: 4/4 coverage AND 0 false alarms.

BNDM / CSDDM / IBDD / OCDD also hit 4/4 — but pay a heavy false-alarm cost.

D3 / SPLL keep 0 FA only by missing half the drift types.

Faster detection on real streams

MEAN TIME TO DETECTION — LOWER IS BETTER



8×

faster · IncrAbrupt 33 vs 272.5

2.4×

faster · IncrRecur 133 vs 323.5

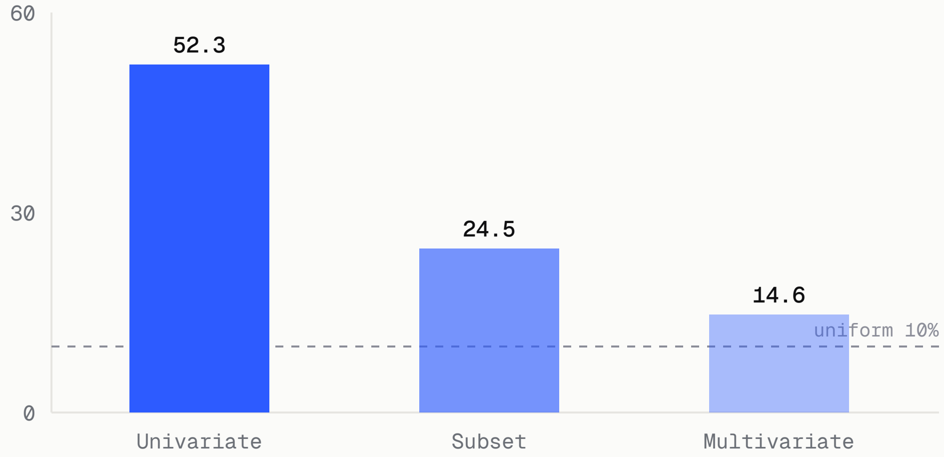
13%

faster · Abrupt 135 vs 155.5

EADD beats D3 on MTD in all three jointly detected INSECTS streams.
InsectsGradual missed by both.

Attribution: P@k = R@k = 1.0

TOP-FEATURE SHAP SHARE (%)

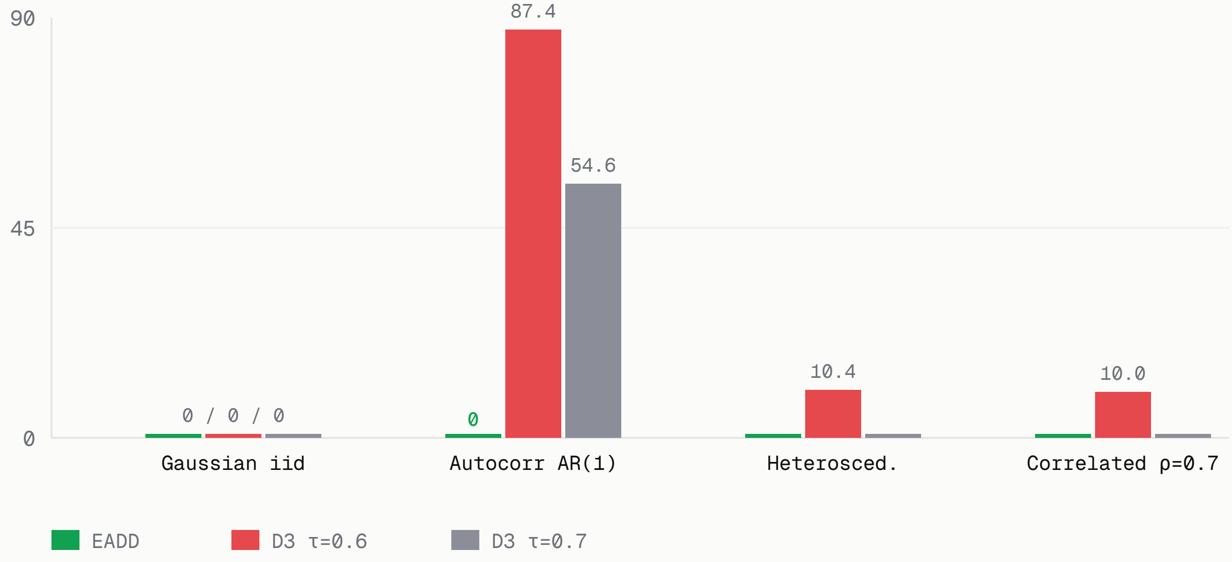


SCENARIO	AUC	TOP FEAT.	SHARE	DSI
Univariate (1/10)	0.784	F3	52.3%	0.193
Subset (3/10)	0.809	F5	24.5%	0.096
Multivariate (10/10)	0.806	F2	14.6%	0.016

DSI orders by concentration: **univariate > subset > multivariate** — consistent with H_{norm}.

Zero false alarms on stable streams

FALSE ALARMS PER STABLE STREAM TYPE — LOWER IS BETTER



0 false alarms on all 4 stable stream types.

Mann–Whitney U vs D3 (τ=0.6): $p = 0.0101$.

ASPT suppresses **54.6** spurious D3 detections on autocorrelated AR(1) at τ=0.7.

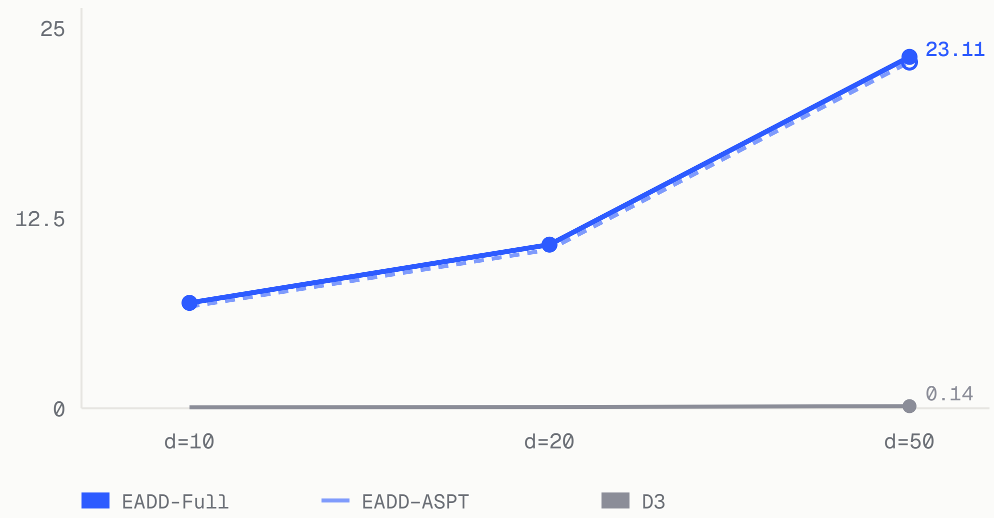
SECTION 04

Ablation & Cost

Where does the gain come from?

Where does the gain come from?

WALL-CLOCK (S) VS DIMENSIONALITY D



Coverage 4/4 vs D3's 2/4 → from LightGBM (non-linear C2ST); EADD–ASPT also reaches 4/4.

0 false alarms on AR(1) → from ASPT; suppresses 54.6 spurious D3 detections.

Cost trade-off ~50–160× D3 wall-clock — dominated by LightGBM, not ASPT (<5%).

05

SECTION 05

Limits & Outlook

What EADD cannot answer — yet.

Limitations & future work

LIMITATIONS

- Wall-clock `~50-160× D3` — LightGBM retrains every `M=50` steps.
- Multiple testing over time (~600 tests / 30K samples). Bonferroni $\alpha' \approx 8.3\times10^{-5}$ is below ASPT's $1/(B_{max}+1) = 0.005$ floor → zero-FA result is empirical, not theoretically guaranteed.
- DSI tiers heuristic — never triggered above LOW in experiments.
- SHAP is correlational, not causal.
- No isolation of reservoir sampling vs window config.

FUTURE WORK

- Anytime-valid **e-values** for streaming FWER.
- Causal attribution — Budhathoki+ AISTATS 2021.
- Multi-domain DSI calibration.
- Adversarial robustness.
- Per-feature operating curves under correlation → feature-group SHAP.

· CONCLUSION

One pipeline, four questions answered

EADD answers WHETHER · WHEN · WHICH · HOW SEVERE — in one pipeline.

4/4	0	8×	1.00
COVERAGE	FALSE ALARMS	FASTER MTD	P@K = R@K

WHAT WE BUILT

- ASPT + H_{pred}: rigorous confirmation, 90–95% permutation savings.
- SHAP + DSI: feature-level diagnosis + severity.
- Benchmark vs 7 unsupervised detectors.

WHAT IT GIVES MLOPS

- Reliable alarms on autocorrelated production streams.
- Actionable diagnoses — not just binary flags.
- Severity-tiered prescriptions.

Thank you.

Questions & discussion welcome.

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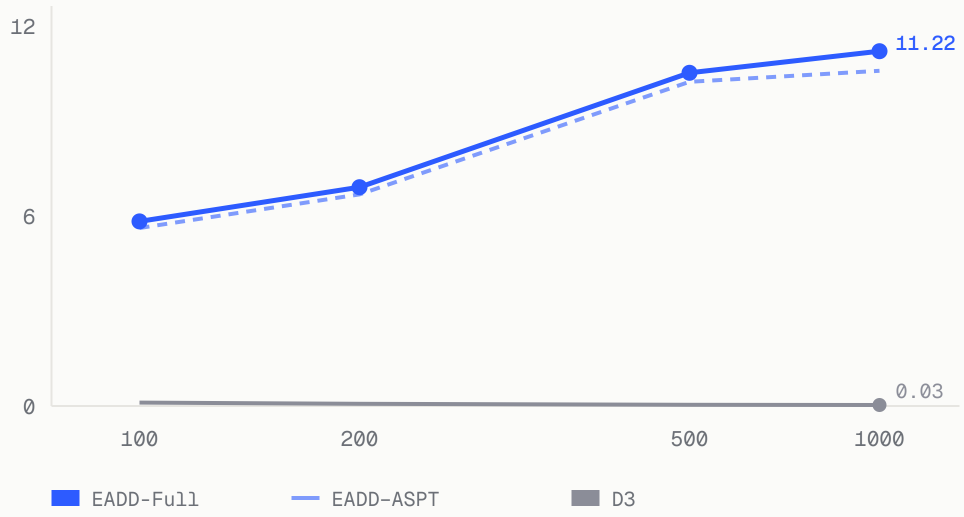
RESOURCES	Code + paper available at the project repository.

Runtime scaling vs window N_{cur}

WALL-CLOCK (S) VS N_{CUR} · LOG-X

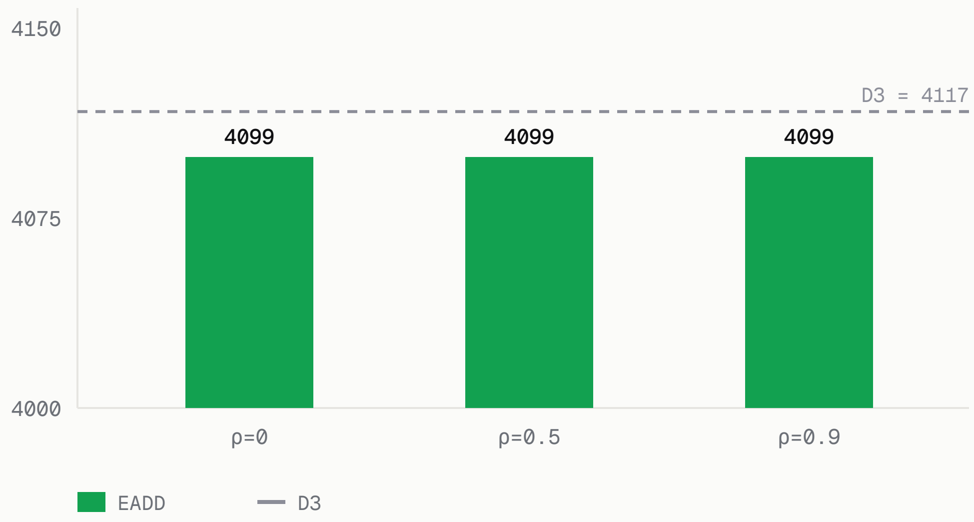


NOTES

- N 200→1000: EADD grows only 1.6×.
- LightGBM sub-linear in N, near-linear in d.
- ASPT overhead <5%.

Correlated features (d=10)

FIRST DETECTION TIME T VS CORRELATION Ρ



NOTES

- Cholesky-mixed Gaussian; +2σ shift in 3 features at `t=4000`.
- EADD robust to correlation; stable correlated streams give `0 FA`.
- SHAP under correlation → interpret at feature-group granularity.